balanced_leverage_max

equity strategy | family: equity-max | status: research | derived from `balanced_leverage`

What this strategy does

balanced_leverage with bigger bets for more return: base unit 0.012, 6x leverage cap

Why it should work. Market-neutral earnings drift with enough gross exposure to matter; the neutrality is the hedge.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.012
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	balanced leverage max	market (CRSP VW)
Annualized return	8.35%	9.94%
Annualized volatility	12.82%	19.35%
Sharpe (NAV-pct)	0.69	0.59
Sortino	0.98	
Max drawdown	-44.9%	-55.5%
Calmar	0.19	
Final NAV ($10M start)	$47,669,815	
Worst day	-7.42%	
% positive days	49.3%	

Sharpe 1996-2001	1.20	
Sharpe 2002-2007	1.39	
Sharpe 2008-2013	-0.41	

Correlation of daily returns with the market: **-0.02**.

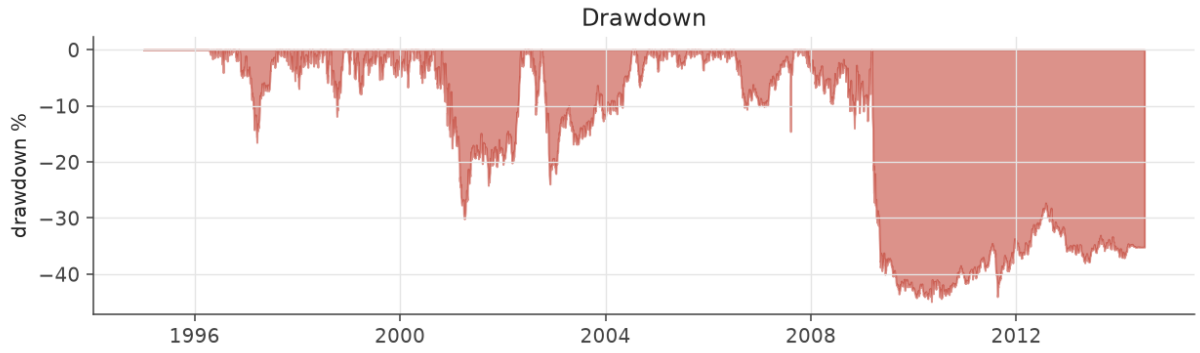
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 0.69

• best return among equity: `market_boost_max` (0.15); this one: 0.0835

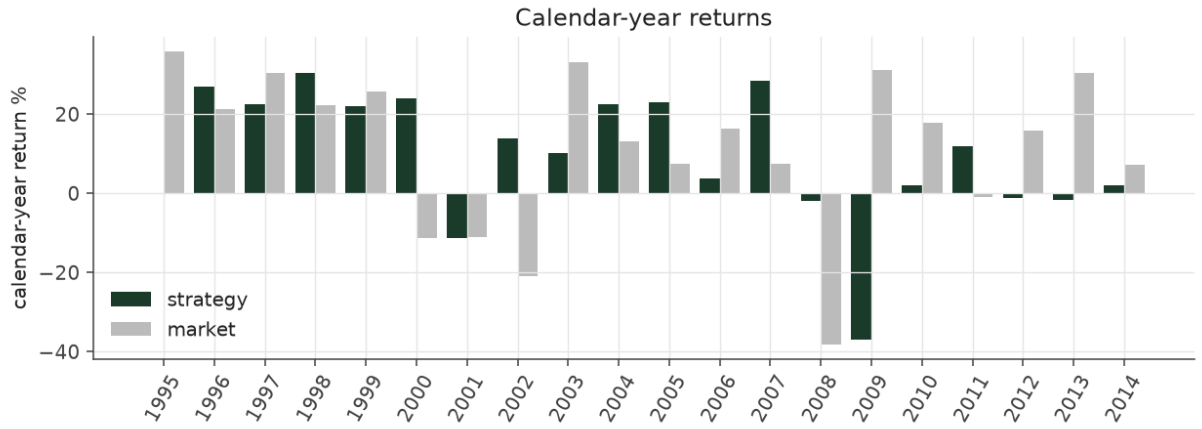
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.186



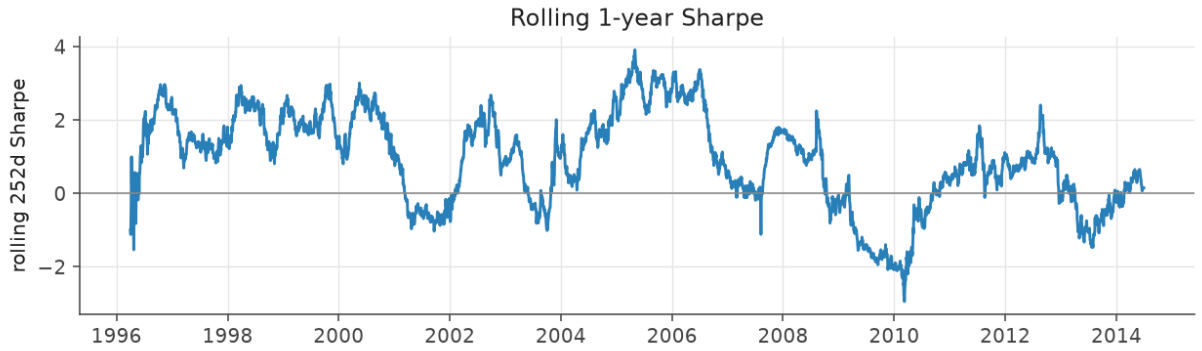
NAV (log scale)



Drawdown from running peak

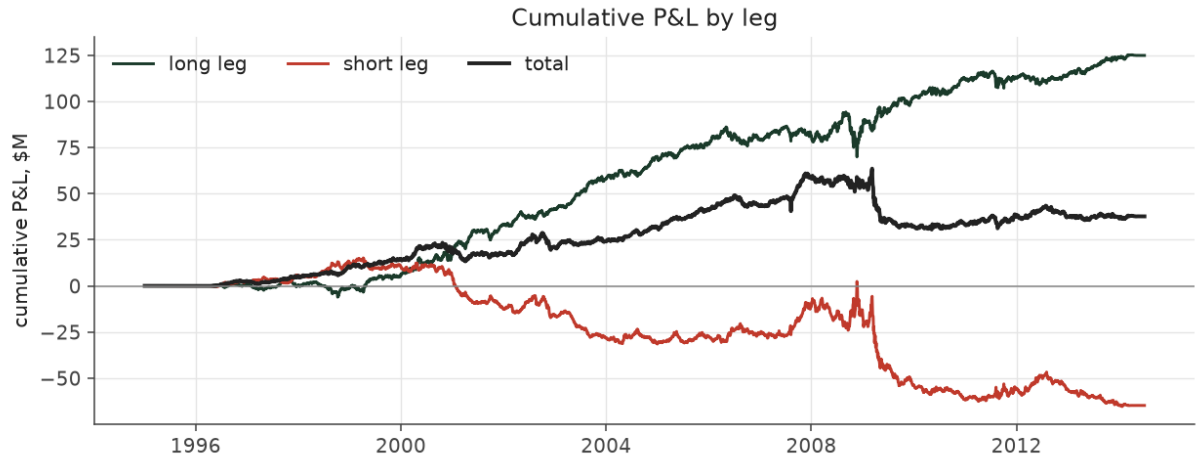


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	124.90	-64.83
P&L after costs ($M)	114.04	-76.37
of which common (average-event) component ($M)	119.86	-122.72

of which excess / PEAD component ($M)	5.05	57.89
excess minus costs ($M)	-5.82	46.35
gross return per $ traded (bp)	177.22	-88.04
positions in the money (hit rate)	0.55	0.48
stand-alone Sharpe (leg only)	1.00	-0.38

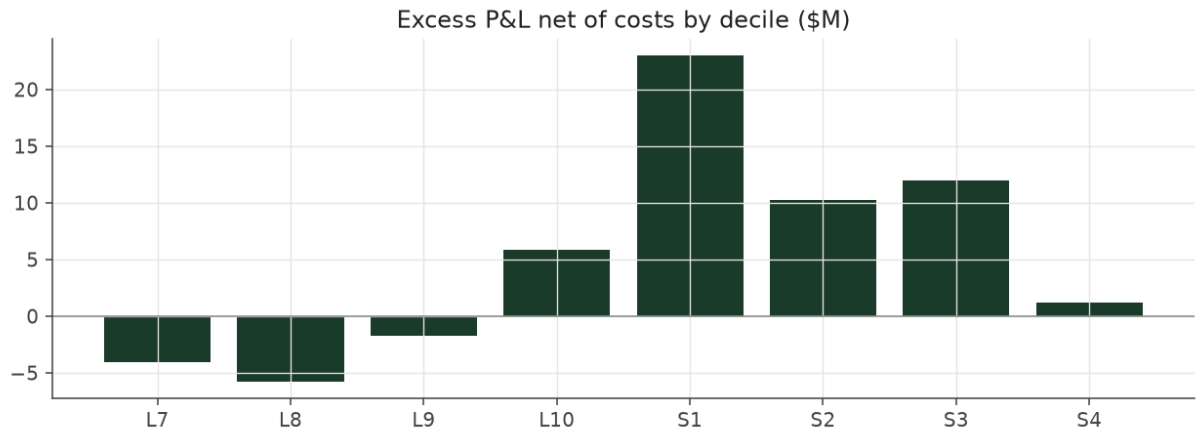
Daily P&L correlation between the two legs: **-0.74**.

• **The short leg loses money in absolute terms** (-64.8M before costs) while the long leg earns +124.9M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 119.9M / 122.7M. Stripping that common component, the surprise signal itself earned +5.0M on the long side and +57.9M on the short side.

• The two legs' daily P&L are -0.74 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.00 long / -0.38 short vs 0.69 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	3,244	418.72	3.15	-3.46	0.66	-4.11	-10.1%	-98.22
long	8	11,112	1,487.56	22.24	-3.49	2.32	-5.81	-14.3%	-39.07
long	9	13,857	2,195.43	32.84	1.63	3.34	-1.70	-4.2%	-7.76
long	10	16,409	2,946.33	55.81	10.36	4.55	5.81	14.3%	19.71
short	1	14,397	2,839.18	-24.90	27.52	4.54	22.98	56.7%	80.96
short	2	14,634	2,617.43	-34.07	14.28	4.07	10.22	25.2%	39.04
short	3	10,525	1,714.05	-15.26	14.62	2.65	11.97	29.5%	69.85
short	4	1,465	192.64	-2.14	1.46	0.28	1.18	2.9%	61.13



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	6,679	331.94	-5.30	-2.98	0.98	-3.97	-9.8%	-119.48
2	14,442	1,582.52	5.65	11.42	3.70	7.72	19.0%	48.77

3	19,426	3,099.11	12.52	18.91	5.81	13.10	32.3%	42.26
4	21,910	4,343.53	13.61	20.01	6.40	13.61	33.6%	31.32
5	23,186	5,054.24	11.20	15.59	5.51	10.08	24.9%	19.94

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	5,125	92.61	0.46	1.35	0.46	0.88	2.2%	95.45
2	14,300	1,181.02	-1.30	4.61	3.54	1.07	2.6%	9.03
3	20,824	3,382.54	18.20	27.98	6.76	21.21	52.3%	62.71
4	22,097	4,695.64	9.39	15.52	6.57	8.95	22.1%	19.05
5	23,293	5,058.34	10.80	13.34	5.06	8.28	20.4%	16.37

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	29,895	5,173.43	-11.36	33.59	7.78	25.80	63.7%	49.88
41-60d	20,346	3,298.60	5.39	21.14	5.64	15.49	38.2%	46.97
61-90d	29,276	5,201.79	34.05	0.54	7.44	-6.90	-17.0%	-13.27
>90d	6,126	737.53	9.58	7.68	1.54	6.14	15.1%	83.22

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	26,452	2,447.71	17.98	21.81	3.85	17.96	44.3%	73.39
2002-07	29,968	5,544.24	39.31	44.52	8.52	36.01	88.8%	64.94
2008-13	29,223	6,419.40	-19.62	-3.39	10.04	-13.44	-33.2%	-20.93

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	13,945	1,413.84	-5.32	-1.40	2.31	-3.71	-9.2%	-26.26
Chems	2,350	911.10	3.53	4.62	1.27	3.35	8.3%	36.80
Durbl	2,464	971.15	-1.63	0.69	1.37	-0.68	-1.7%	-6.99
Enrgy	5,523	1,239.10	10.99	13.53	1.79	11.73	29.0%	94.70
Hlth	8,778	1,529.54	8.27	10.27	2.44	7.83	19.3%	51.20
Manuf	9,286	1,371.31	1.37	3.28	2.21	1.07	2.6%	7.81
Money	9,738	1,006.41	3.77	5.27	1.76	3.51	8.7%	34.90
NoDur	4,402	1,165.92	5.68	6.90	1.73	5.17	12.8%	44.34
Other	11,612	1,284.04	6.67	9.27	2.18	7.09	17.5%	55.22
Shops	9,830	1,499.65	9.11	11.15	2.42	8.73	21.5%	58.20

Telcm	3,469	1,055.51	-4.45	-2.45	1.45	-3.90	-9.6%	-36.98
Utils	4,246	963.79	-0.32	1.80	1.46	0.34	0.8%	3.49

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	29,591	5,215.15	-10.61	1.87	7.58	-5.71	-14.1%	-10.95
low vol	27,167	4,306.69	25.27	33.04	7.18	25.86	63.8%	60.05
mid vol	27,433	4,712.18	21.91	26.96	7.34	19.63	48.4%	41.65

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	8.35%	0.00%
0.5%	7.1	7.46%	0.89%
1.0%	14.1	6.41%	1.94%
2.0%	28.3	3.46%	4.89%
5.0%	70.7	-100.00%	108.35%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.